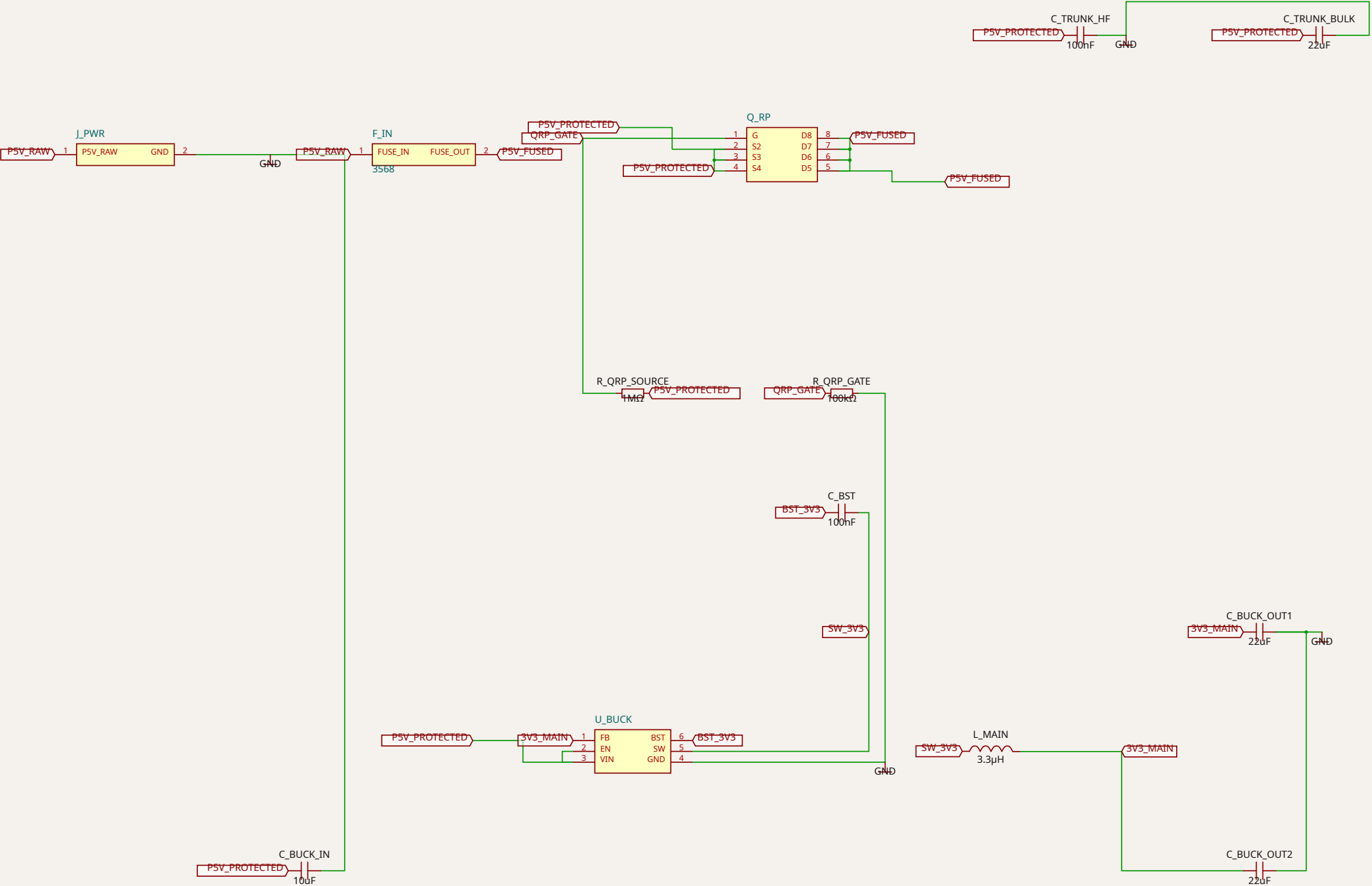


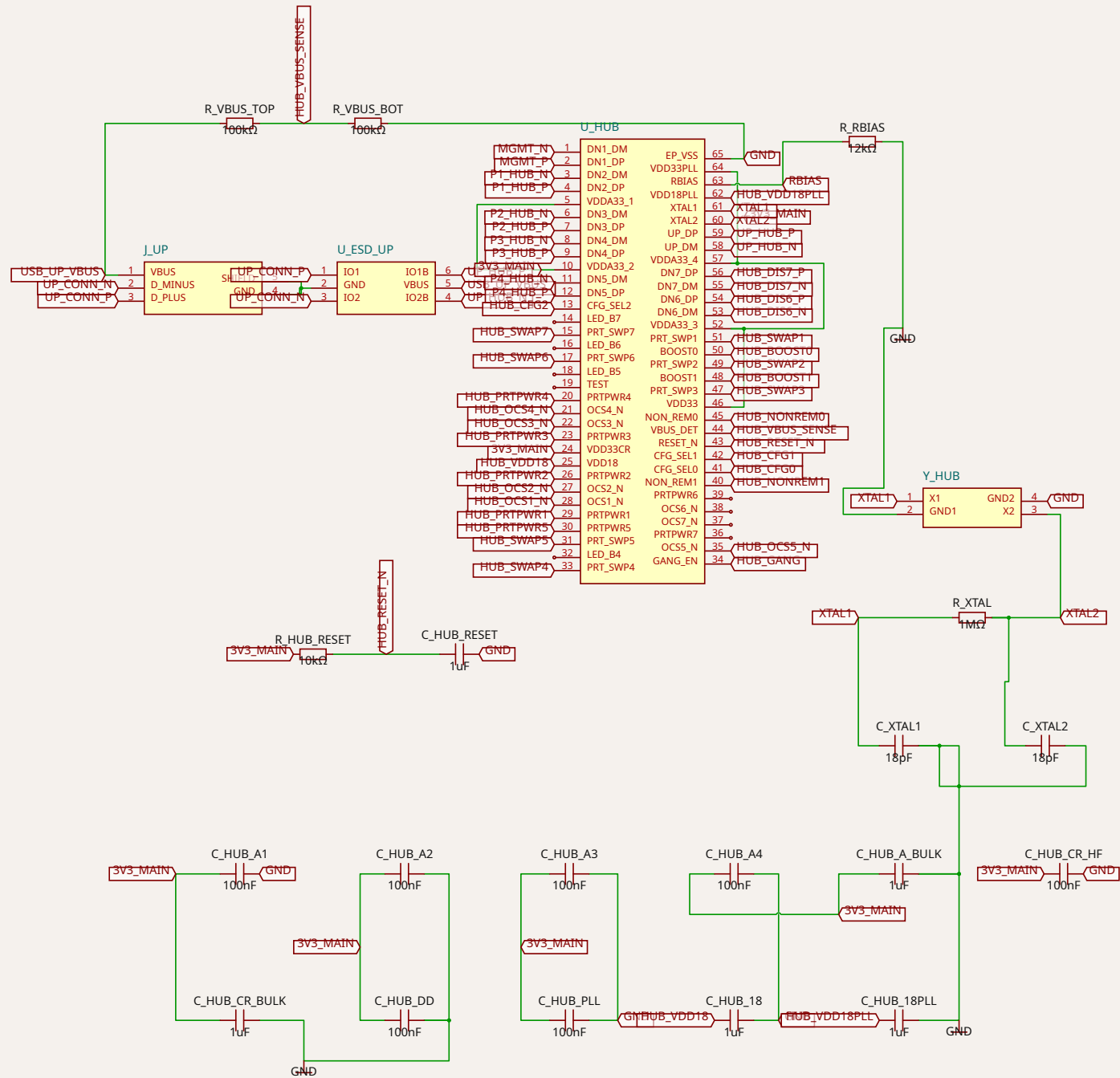
USB_CONTROLLED_DEBUG_HUB

POWER — regulated 5 V input, fuse, reverse protection and 3.3 V buck

Page 1 of 9 • exact circuit.json SHA-256 dff07e17269025a4... • 13 components • landscape page fit



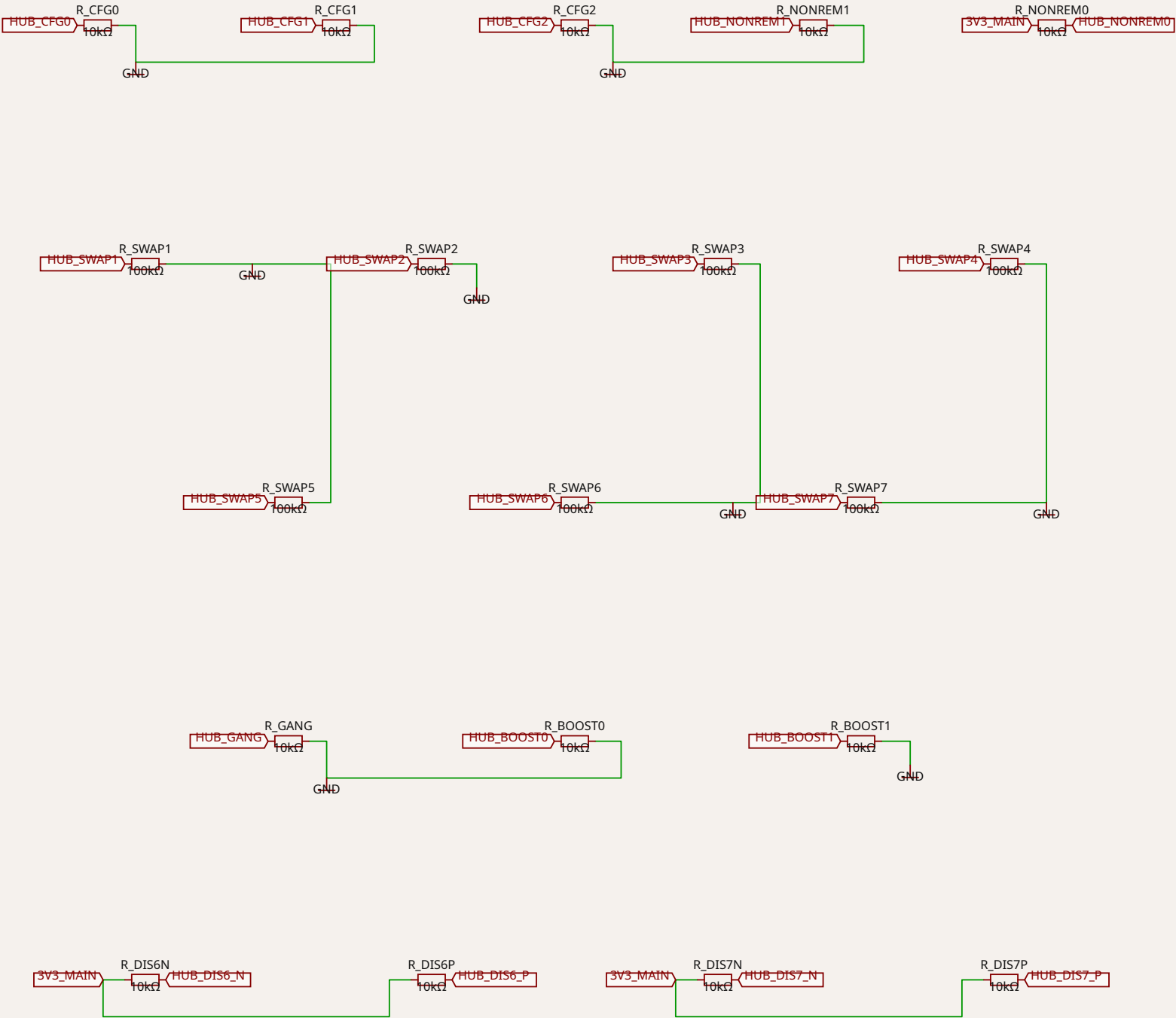
USB HUB — upstream Type-B, ESD, USB2517I, straps, clock and bypass



USB_CONTROLLED_DEBUG_HUB

USB HUB CONFIGURATION — factory-default straps and disabled physical ports 6/7

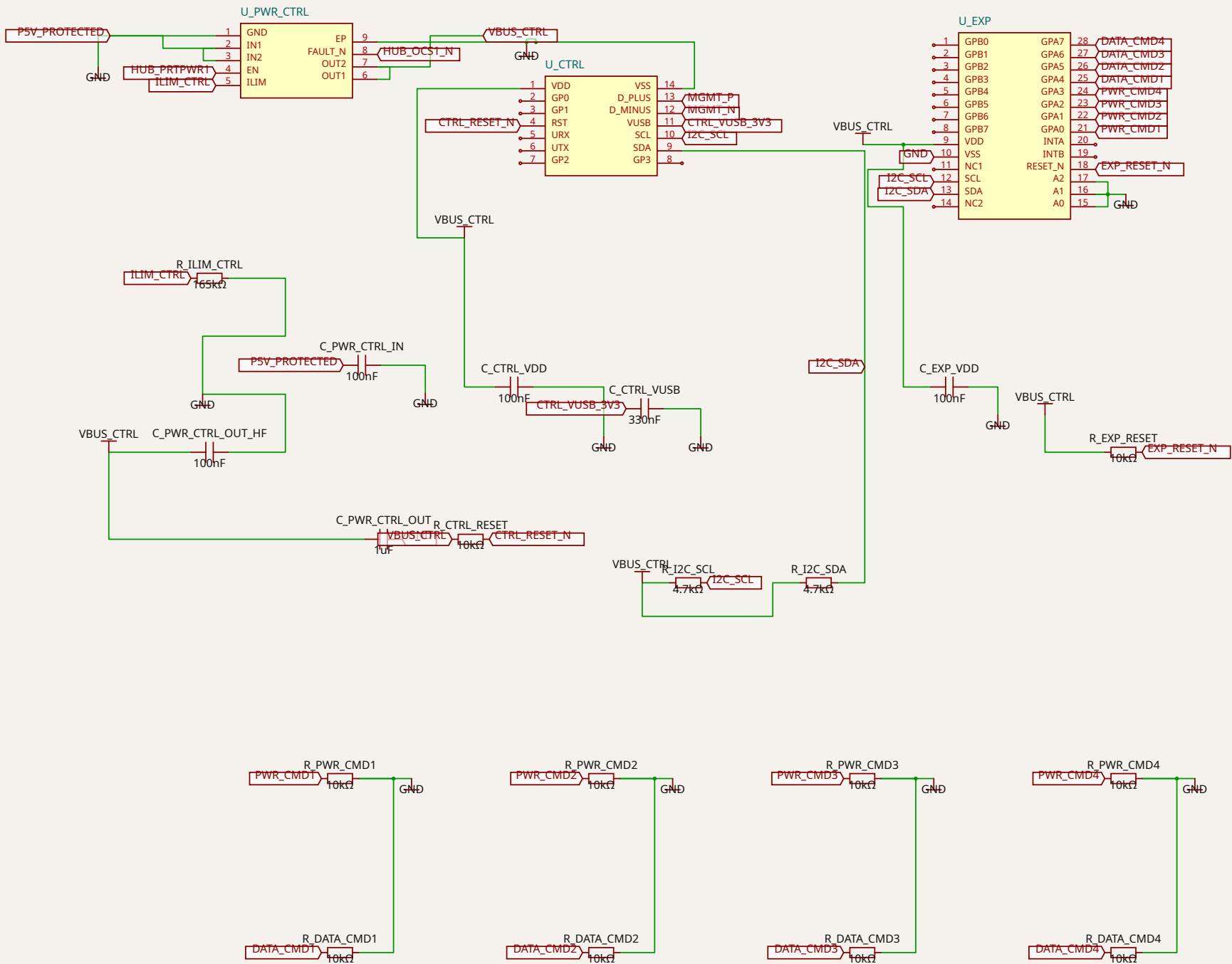
Page 3 of 9 • exact circuit.json SHA-256 dff07e17269025a4... • 19 components • landscape page fit



USB_CONTROLLED_DEBUG_HUB

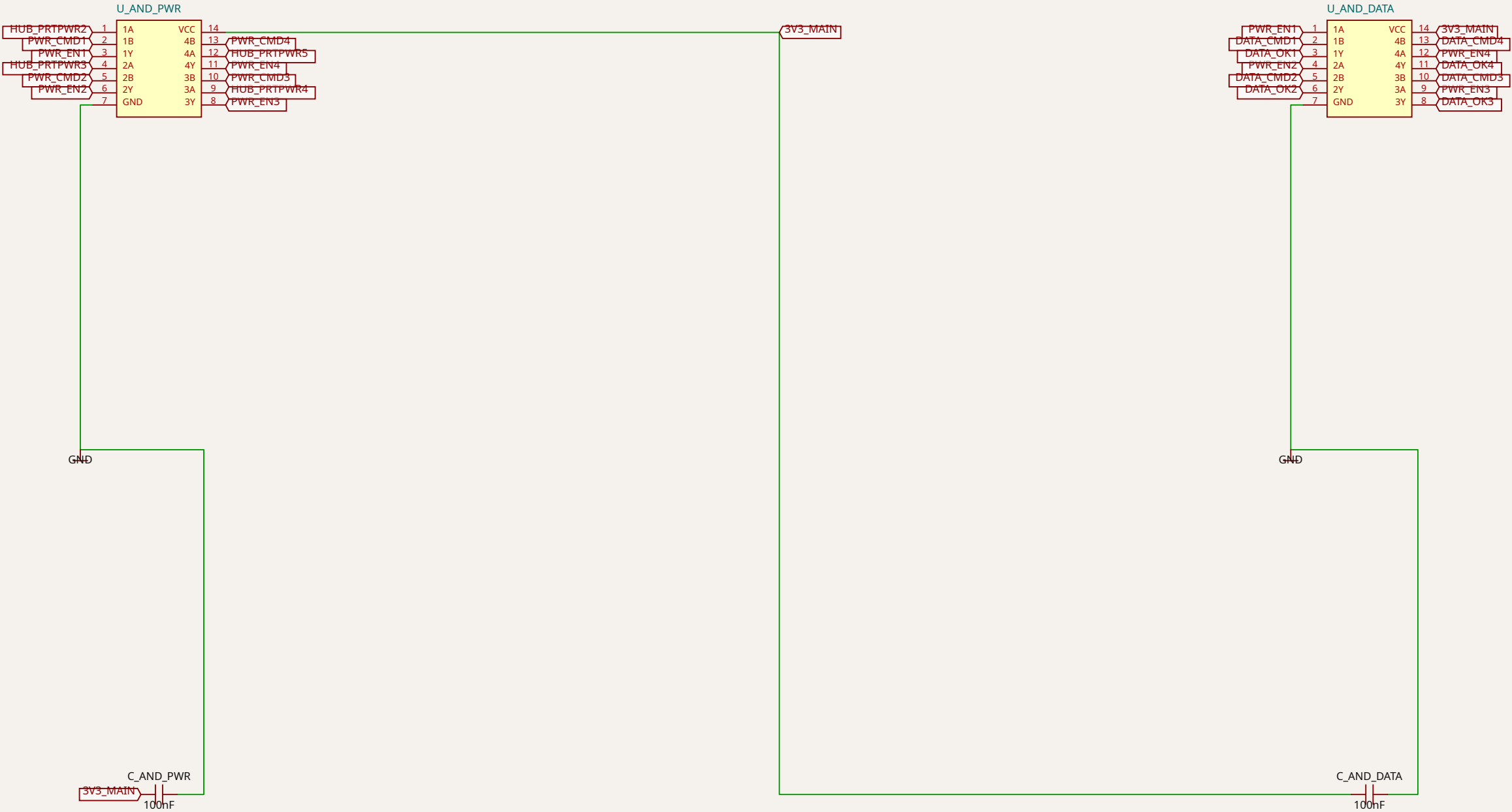
MANAGEMENT — factory MCP2221A HID/I2C and MCP23017 command bank

Page 4 of 9 • exact circuit.json SHA-256 dff07e17269025a4... • 22 components • landscape page fit



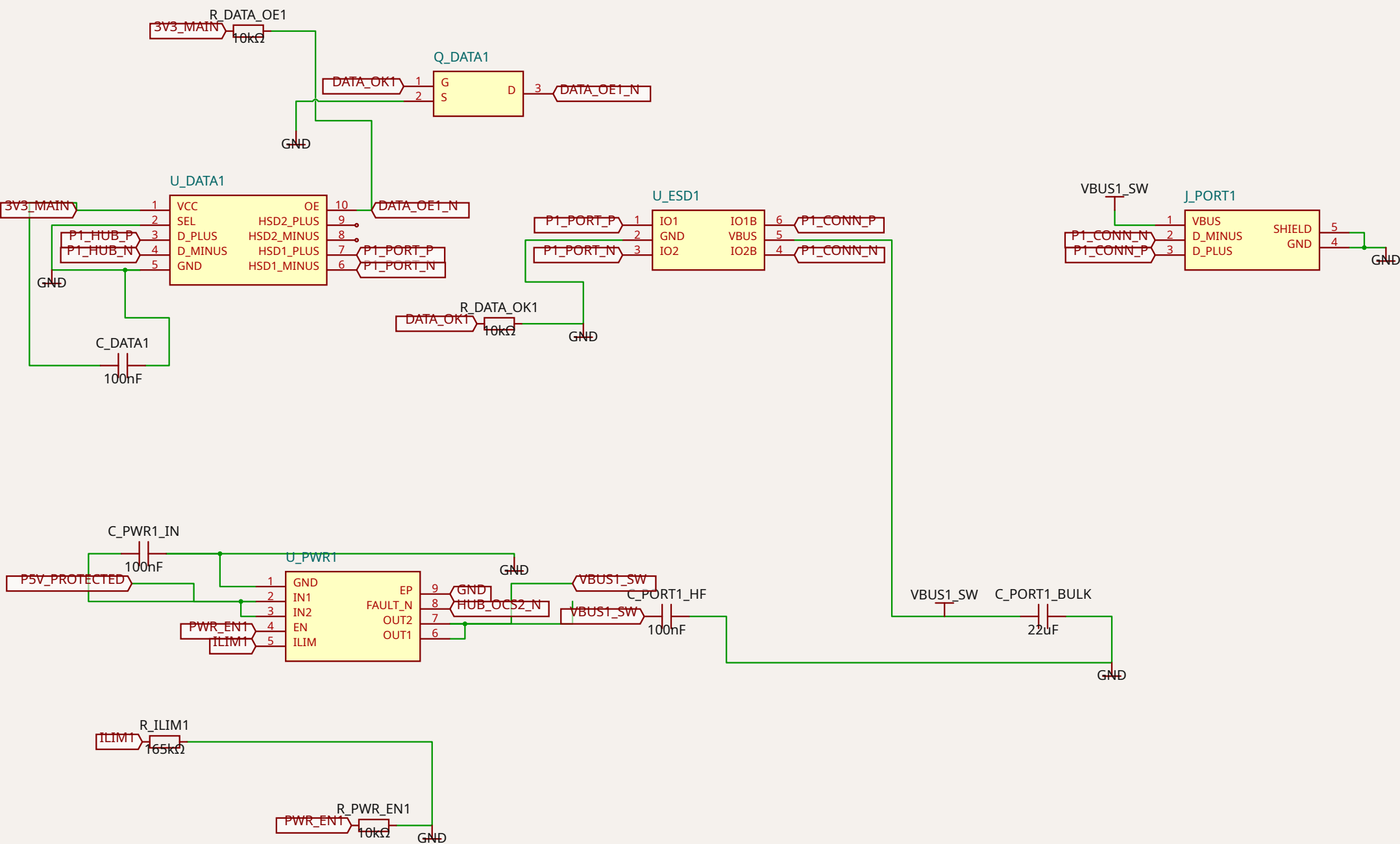
USB_CONTROLLED_DEBUG_HUB

INTERLOCKS — hub policy AND host command; data requires final power enable



USB_CONTROLLED_DEBUG_HUB

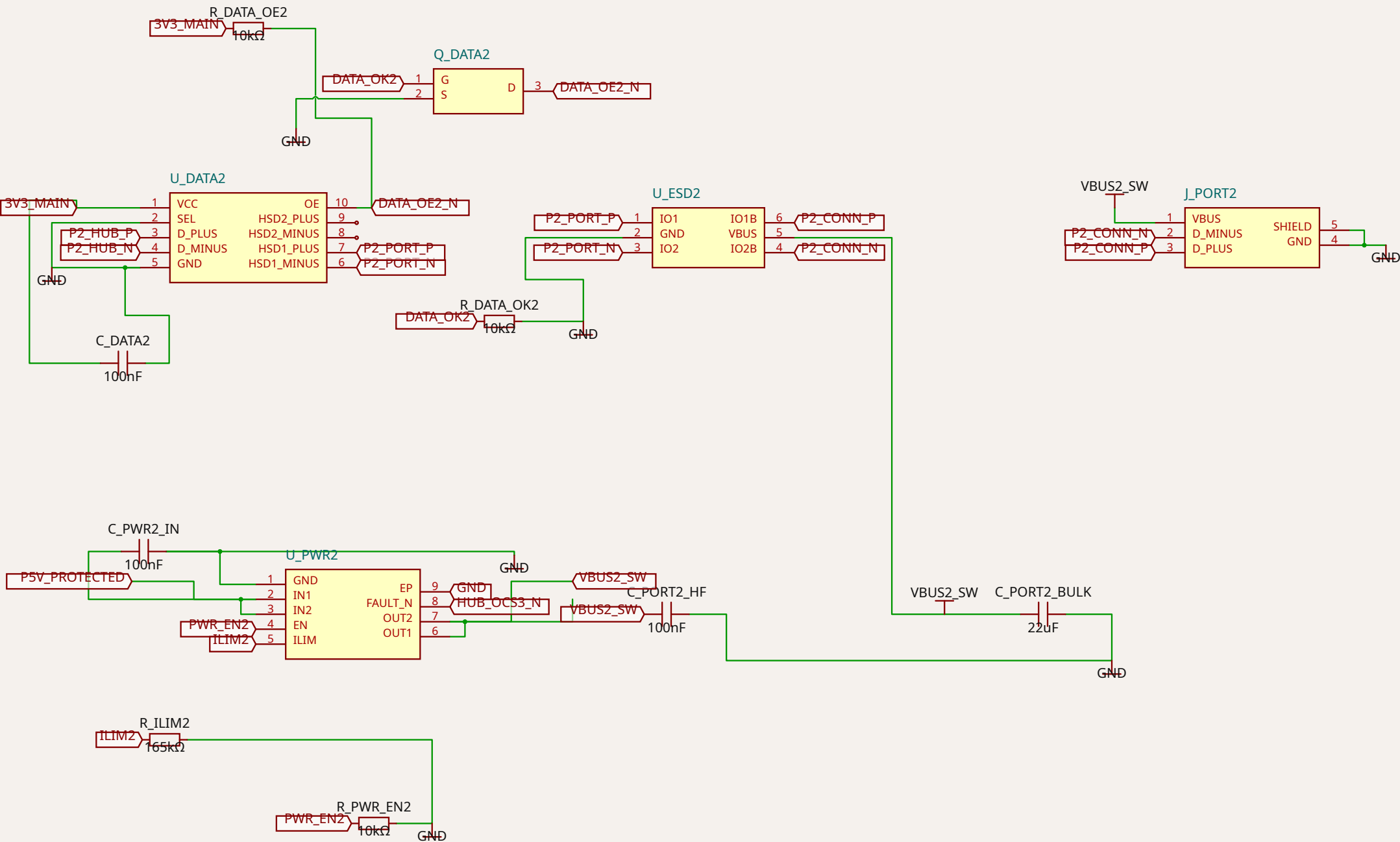
EXTERNAL PORT 1 — independent power/data disconnect



USB_CONTROLLED_DEBUG_HUB

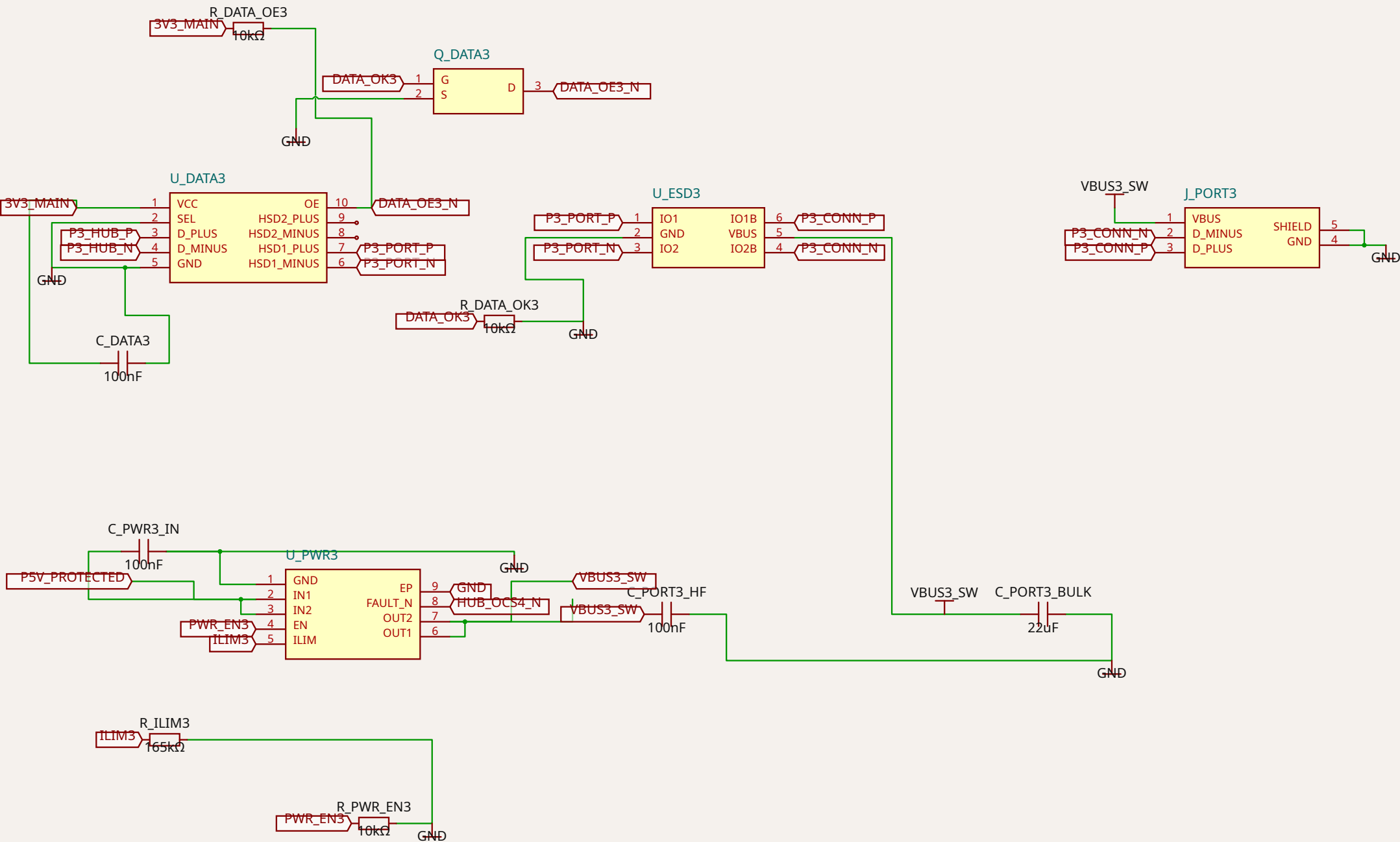
EXTERNAL PORT 2 — independent power/data disconnect

Page 7 of 9 • exact circuit.json SHA-256 dff07e17269025a4... • 13 components • landscape page fit



USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 3 — independent power/data disconnect



USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 4 — independent power/data disconnect

